

Worksheet

1. Compute $\langle 7, 15 \rangle + 3 \langle -1, 2 \rangle$.
2. Compute $\langle 3, 5, -2 \rangle - 4 \langle -1, 0, 3 \rangle$.
3. Find $\|2\mathbf{i} - 3\mathbf{j} + \mathbf{k}\|$.
4. Find a unit vector that points in the same direction as $\langle 5, -1, 4 \rangle$.
5. A force of $2\mathbf{i} - 3\mathbf{j} + \mathbf{k}$ Newtons and a force for $-\mathbf{i} - 5\mathbf{j} + 3\mathbf{k}$ Newtons act on a particle. Find the net force on the particle and find a unit vector acting in the same direction.
6. Write $x^2 - 4x + y^2 + 5y + z^2 - 6z = 5$ in the standard form for a sphere and find its center and radius.
7. Why do you think a scalar is called a “scalar”? What is getting “scaled”?